10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 24 — Carboxylic Acids

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

24.1 Can you oxidize methanol with $KMnO_4$ just like ethanol?

Yes. Hot acidified potassium permanganate is a very efficient oxidising agent, and it oxidises methanol in the same way as ethanol. Methanol is first oxidised to methanal and then to methanoic acid. In this reaction the purple colour of potassium permanganate changes to colourless, because the manganese of $KMnO_4$ is reduced.

$$CH_3OH + [O] \rightarrow HCHO + H_2O$$

$$HCHO + [O] \rightarrow HCOOH$$

Here [O] shows the oxygen supplied by the acidified $KMnO_4$.

24.1 Explain the difference between a fermentation reaction and a chemical reaction.

A fermentation reaction is also a chemical reaction, but it is carried out by living cells. The difference is that fermentation needs microorganisms (bacteria or yeast) and their enzymes, whereas an ordinary chemical reaction is carried out with chemical reagents in the laboratory or industry.

Fermentation reaction	Chemical reaction
It is carried out by microorganisms such as yeast or acetic acid bacteria.	It is carried out by chemical reagents, and a catalyst may also be used.
The enzymes of the microorganisms act as catalysts.	Heat, pressure, light or a chemical catalyst is used.
It takes place under mild conditions, usually at about 35 °C and ordinary pressure.	The conditions depend on the reaction; strong heating or high pressure may be needed.
Example: glucose gives ethanol and CO_2 in the presence of yeast.	Example: ethanol is oxidised to ethanoic acid by hot acidified $KMnO_4$.

24.2 Is ethanoic acid a weak acid?

Yes. Ethanoic acid is a weak acid as compared to the mineral acids, because it does not give its proton (H^+) easily. It ionises only partly in water, so a small number of hydrogen ions are produced and the solution is only weakly acidic. That is why it reacts slowly with metals and carbonates.

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

24.2 How will calcium carbonate react with ethanoic acid?

Calcium carbonate reacts with ethanoic acid to give calcium ethanoate, carbon dioxide and water. Effervescence takes place because carbon dioxide gas escapes from the solution, and calcium ethanoate remains dissolved in the water.

$$2CH_3COOH + CaCO_3 \rightarrow (CH_3COO)_2Ca + CO_2 + H_2O$$

24.3 Explain the function of a preservative.

A preservative is a chemical compound which is added to foodstuff to prevent or retard the growth of microorganisms. In this way it stops the food from spoiling, maintains its quality and extends its shelf-life. Benzoic acid, acetic acid, propionic acid and the methyl and ethyl esters of 4-hydroxy benzoic acid are common food preservatives, and acetic acid is very widely used as a preservative in pickles.

24.3 Give examples of some citrus fruits.

Lemons, oranges, limes and grapefruit are citrus fruits. They contain citric acid, which gives them their sour taste.

Exercise (Solved)

A. Multiple Choice Questions

1. Identify the ester functional group:

(a) –COOR (b) –COOH

(c) =C=O (d) $–CH_2–OH$

Answer: (a) — –COOR is the ester group, in which the –OH of the carboxyl group is replaced by an –OR group. –COOH is the carboxyl group and $–CH_2OH$ is the alcoholic group.

2. During the oxidation of ethanol by acidified $KMnO_4$, which element is reduced?

(a) K (b) O

(c) Mn (d) H

Answer: (c) — In $KMnO_4$ the manganese is in the +7 oxidation state. During the reaction it is reduced to Mn^{2+} (colourless), which is why the purple colour of the reagent disappears.

3. Fermentation of foodstuff in the presence of acetic acid bacteria needs a sufficient supply of oxygen. What is the role of oxygen in this process?

(a) It enhances the reactivity of bacteria (b) It oxidizes ethanol which is formed during the reaction

(c) It inhibits the destruction of bacteria (d) It inhibits the oxidation of acetic acid

Answer: (b) — The bacteria convert the alcohol of the foodstuff into vinegar, and oxygen is the oxidising agent which oxidises this ethanol to ethanoic acid. That is why air is bubbled through the liquid and the liquid is stirred continuously.

4. Which organic acid is used as a component of food?

(a) Formic acid (b) Oxalic acid
(c) Propanoic acid (d) Acetic acid

Answer: (d) — Acetic acid is the acid of vinegar, so it is used as a component of food and as a preservative in pickles and sauces.

5. Which of the following acids is present in apples?

(a) Citric acid (b) Tartaric acid
(c) Acetic acid (d) Malic acid

Answer: (d) — Malic acid is present in apples, grapes and carrots, while citric acid is present in lemons and other citrus fruits.

6. Which substances will react to produce CH_3–CH_2–CH_2–$COOC_2H_5$?

(a) $CH_3CH_2CH_2COOH$ and CH_3OH (b) CH_3CH_2COOH and CH_3CH_2OH
(c) $CH_3CH_2CH_2COOH$ and CH_3CH_2OH (d) $CH_3CH_2CH_2CH_2COOH$ and CH_3OH

Answer: (c) — The ester is made of an acid part and an alcohol part. The acid part $CH_3CH_2CH_2CO$– comes from butanoic acid and the alkyl part –OC_2H_5 comes from ethanol, so butanoic acid and ethanol are the reactants; the ester is ethyl butanoate.

7. What is the formula of ester formed by the reaction of methanol with propanoic acid?

(a) $CH_3COOCH_2CH_3$ (b) $CH_3CH_2COCH_3$
(c) $CH_3CH_2COOCH_3$ (d) $CH_3CH_2OCOCH_3$

Answer: (c) — Propanoic acid, CH_3CH_2COOH, loses –OH and methanol, CH_3OH, loses H from its –OH group, so the ester formed is methyl propanoate, $CH_3CH_2COOCH_3$.

8. Which polymer is made by reacting a diol and a dicarboxylic acid?

(a) Polyethene (b) Polyester
(c) Polystyrene (d) Polyamide

Answer: (b) — A very important polymer, polyester, is made by reacting diols with dicarboxylic acids. Polyester is used to make synthetic fibres, films and bottles.

9. Which compound is used as a synthetic fiber?

(a) Methyl acetate (b) Polyester
(c) Resin (d) n-Amyl acetate

Answer: (b) — Esters have an important role in the production of synthetic polymers, and polyester is the ester used as a synthetic fibre. Methyl acetate and n-amyl acetate are used as solvents and flavouring agents.

studysapiens.com

B. Short Answer Questions

24.1 What is an esterification reaction?

Carboxylic acids form esters when they are heated with excess of an alcohol in the presence of concentrated H_2SO_4 as a catalyst. This reaction is called the esterification reaction. In this reaction the –OH group of the acid and the H atom of the alcohol combine to form a molecule of water, so an ester and water are obtained. Esters have pleasant fruity smells and are used as flavours and fragrances.

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O \quad (\text{conc. } H_2SO_4, \text{ heat})$$

24.2 Name an oxidizing agent used to oxidize alcohols.

Hot acidified potassium permanganate ($KMnO_4$) is used as an oxidising agent to oxidize alcohols. Acidified potassium dichromate ($K_2Cr_2O_7$) can also be used for this purpose.

24.3 Why sufficient oxygen is provided during fermentation to prepare acetic acid from ethanol?

Acetic acid bacteria can produce vinegar from alcoholic foodstuffs only when sufficient oxygen is provided, because oxygen oxidises the ethanol into ethanoic acid. For this reason oxygen (air) is bubbled through the liquid in the fermentation tank and the liquid is stirred continuously. In this way vinegar containing about 15 % ethanoic acid is prepared in only 24 hours.

$$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O \quad (\text{acetic acid bacteria})$$

24.4 What is the role of carboxylic acids in the preparation of beverages?

Carboxylic acids such as tartaric acid, citric acid and malic acid are used as flavour enhancers in the food industry. The sour taste of foods and beverages is largely due to these acids, and they improve the taste of soft drinks and other beverages. Some acids also act as preservatives, so they keep the beverage usable for a longer time.

24.5 Give a practical example in which an acid or an ester is used as a flavouring agent.

Citric acid is used as a flavouring agent in soft drinks and gives them a pleasant sour taste. Among the esters, ethyl butyrate is used to give a pineapple flavour, methyl butyrate is used for an apple flavour and n-Octyl acetate is used for an orange flavour.

24.6 Which carboxylic acids are very often used as flavour enhancers?

Tartaric acid, citric acid and malic acid are very commonly used as flavour enhancers in the food industry. Acetic acid is also used, because it gives vinegar its characteristic taste.

24.7 Which property of acetic acid does make it suitable to act as preservative in pickle?

Acetic acid is a weak acid, so it lowers the pH of the pickle and makes the medium acidic. In this acidic medium the growth of spoilage microorganisms is prevented or retarded, so the pickle does not spoil and its shelf-life increases. Acetic acid is therefore very widely used as a preservative in pickles.

C. Constructed Response Questions

24.1 How does ethanol react with propanoic acid?

When ethanol is heated with propanoic acid in the presence of a few drops of concentrated sulphuric acid, esterification takes place. The –OH group of propanoic acid and the H atom of the alcohol combine to form water, so ethyl propanoate and water are formed. Ethyl propanoate is an ester and it has a pleasant fruity smell. The concentrated H_2SO_4 acts as a catalyst and also absorbs the water formed, so the reaction goes to the right.

$$CH_3CH_2COOH + C_2H_5OH \rightleftharpoons CH_3CH_2COOC_2H_5 + H_2O$$

24.2 How does acetic acid react with Mg and $MgCO_3$?

With magnesium: dilute ethanoic acid reacts with magnesium metal to give magnesium ethanoate and hydrogen gas. The reaction is rather slow because carboxylic acids are weaker than the mineral acids, and bubbles of hydrogen gas are seen.

$$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$$

With magnesium carbonate: effervescence takes place because carbon dioxide gas is evolved, and magnesium ethanoate is formed in the solution along with water.

$$2CH_3COOH + MgCO_3 \rightarrow (CH_3COO)_2Mg + CO_2 + H_2O$$

24.3 How are carboxylic acids used as solvents?

Acetic acid dissolves many compounds such as oils, sulphur, iodine and water, so in industry it is used as a solvent for resins, paints and lacquers. Acetic acid is also called glacial acetic acid; it is a corrosive, colourless solid which is completely miscible with water. Esters are also commonly used as solvents: ethyl acetate is used to extract caffeine from coffee and as nail polish and paint remover, while other volatile esters are used as solvents for coatings, paints, varnishes, plastics, resins and lacquers.

24.4 How are carboxylic acids used in daily life?

(i) In food and drinks: acetic acid is used in vinegar, pickles and sauces, while citric acid, tartaric acid and malic acid are used as flavour enhancers in soft drinks and sweets.

(ii) As preservatives: benzoic acid, acetic acid and propionic acid are added to jams, juices, milk and other drinks to prevent the growth of microorganisms and to extend their shelf-life.

(iii) In soaps and cosmetics: palmitic acid and stearic acid are used to make soaps, cosmetics, candles and protective coatings, while oleic acid is used in soaps and detergents.

(iv) In medicines and plastics: carboxylic acids such as aspirin and ibuprofen are used as medicines, acetic acid is used to make cellulose acetate plastic, and acrylic acid is used to prepare long-chain polymers called acrylates.

D. Descriptive Questions

24.1 Explain fermentation reaction which is used to prepare acetic acid on commercial scale.

Alcoholic foodstuffs such as grain, malt, rice or potato mesh are converted into vinegar when they undergo fermentation in the presence of acetic acid bacteria. These bacteria can produce vinegar from alcoholic foodstuffs if sufficient oxygen is provided, because oxygen oxidises the alcohol into acetic acid.

$$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O \quad \text{(acetic acid bacteria)}$$

On the commercial scale, the alcoholic liquid is placed in a fermentation tank which is provided with a continuous supply of oxygen. The oxygen is bubbled through the liquid and the liquid is stirred continuously, so that the bacteria get enough oxygen and all parts of the liquid are exposed to them. In this way vinegar containing about 15 % ethanoic acid is prepared in only 24 hours. The product is then filtered and diluted to the concentration required for table vinegar. This process is different from the fermentation of glucose, because that reaction takes place in the absence of oxygen, while this oxidative fermentation needs a continuous supply of oxygen.

24.2 Describe the industrial applications of carboxylic acids and esters.

(i) As solvents: acetic acid dissolves oils, sulphur, iodine and water, so it is used as a solvent for resins, paints and lacquers. Ethyl acetate is used to extract caffeine from coffee and as nail polish and paint remover, and other volatile esters are used as solvents for coatings, varnishes, plastics and resins.

(ii) As flavouring agents: tartaric acid, citric acid and malic acid are used as flavour enhancers in the food industry, and esters with pleasant odours are added to food products to improve their smell and taste.

(iii) In synthetic fibres and plastics: polyester is made by reacting diols with dicarboxylic acids, and acetic acid is used in the production of cellulose acetate plastic. Acrylic acid is a starting reagent to prepare long-chain polymers called acrylates, and polymethyl methacrylate (Plexiglass) is used as a substitute of glass.

(iv) In soaps, cosmetics and medicines: palmitic and stearic acids are used in soaps, cosmetics, pharmaceuticals, candles and protective coatings, oleic acid is used in soaps and detergents, and acids such as aspirin and ibuprofen are used as medicines. Citric acid esters are used as plasticizers in many industries.

24.3 What are flavouring agents? Which carboxylic acids are used as flavouring agents?

Flavouring agents are substances which are added to give an additional taste or flavour to a substance. They also help in masking the unpleasant taste of drugs, especially those used by children. Organic acids such as tartaric acid, citric acid and malic acid are very commonly used as flavour enhancers in the food industry. The sour taste of foods is largely due to these acids, and they improve the taste of food and beverage products, especially soft drinks. Acetic acid gives vinegar its characteristic taste and is used in pickles and sauces.

Ester	Flavour
Methyl butyrate	Apple
Ethyl butyrate	Pineapple
n-Amyl acetate	Pears, bananas
n-Octyl acetate	Orange
Methyl salicylate	Apricot

24.4 Explain the reactions of carboxylic acids with metals and carbonates.

Reaction with metals: carboxylic acids are much weaker acids than the mineral acids, so they react with more reactive metals rather slowly and give a salt with the evolution of hydrogen gas. For example, dilute ethanoic acid reacts with magnesium metal to give magnesium ethanoate and hydrogen gas.

$$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$$

Reaction with carbonates: when solid sodium carbonate is mixed with an aqueous solution of ethanoic acid, effervescence takes place due to the evolution of carbon dioxide gas and sodium ethanoate is formed in the solution. In the same way calcium carbonate gives calcium ethanoate, carbon dioxide and water.

$$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$$
$$2CH_3COOH + CaCO_3 \rightarrow (CH_3COO)_2Ca + CO_2 + H_2O$$

The carbon dioxide which comes out turns lime water milky, and this is used as a test for the carbonate.

24.5 Describe applications of esters in daily life.

(i) As flavours and fragrances: esters have pleasant smells and are responsible for the flavours of many fruits, so they are used as flavouring agents and fragrances in food products, perfumes and cosmetics.

(ii) As solvents: ethyl acetate is used to extract caffeine from coffee and is used as nail polish and paint remover, while other volatile esters are used as solvents for coatings, paints, varnishes, plastics and lacquers.

(iii) In fibres and plastics: polyester is made by reacting diols with dicarboxylic acids and is used to make synthetic fibres, films and bottles, and citric acid esters are used as plasticizers in many industries.

(iv) In soaps and preservatives: esters are important in the manufacture of soaps and detergents, and the methyl and ethyl esters of 4-hydroxy benzoic acid are used as food preservatives. Fats and oils are also esters and are the raw materials for soap manufacture.

studysapiens.com